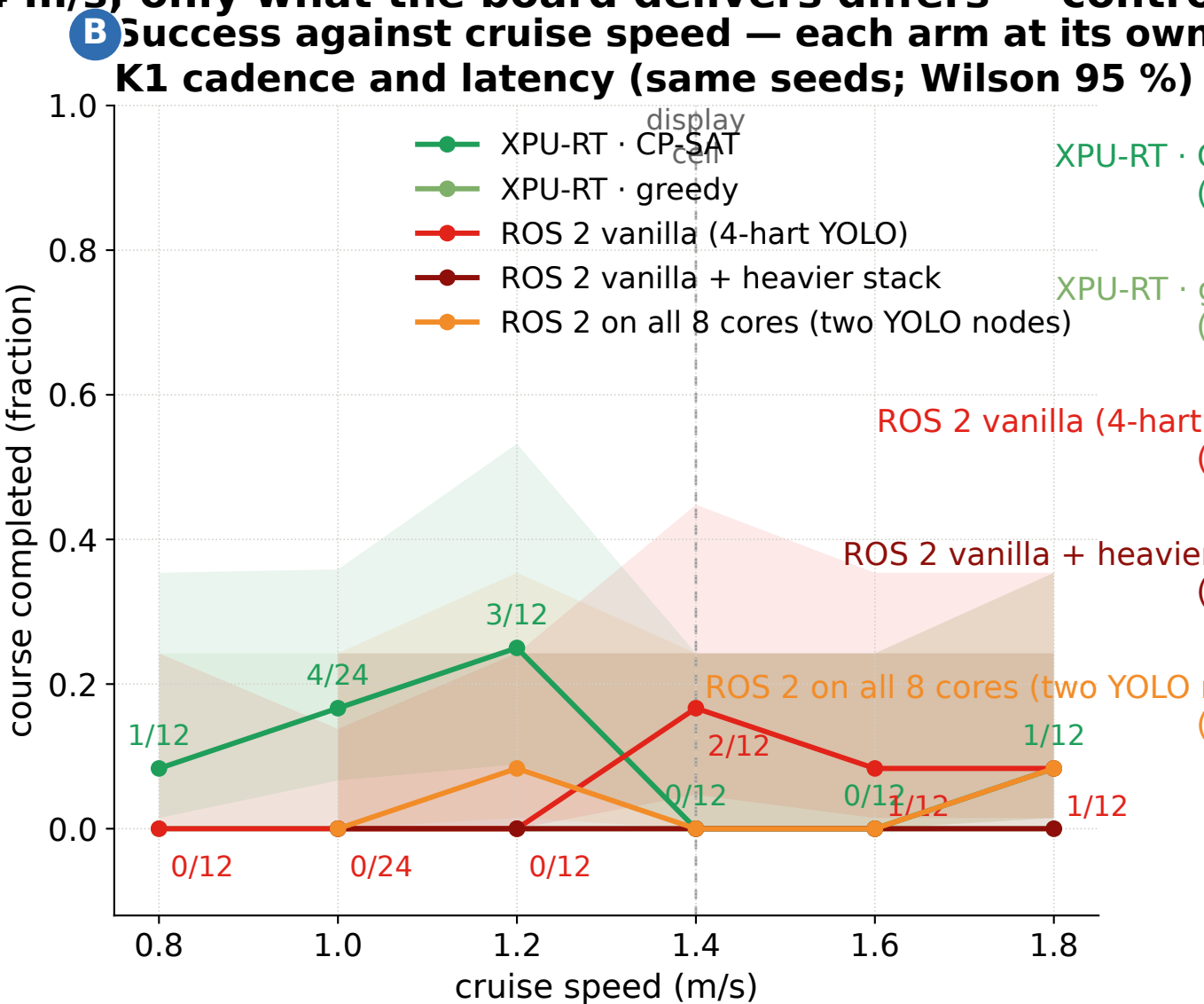
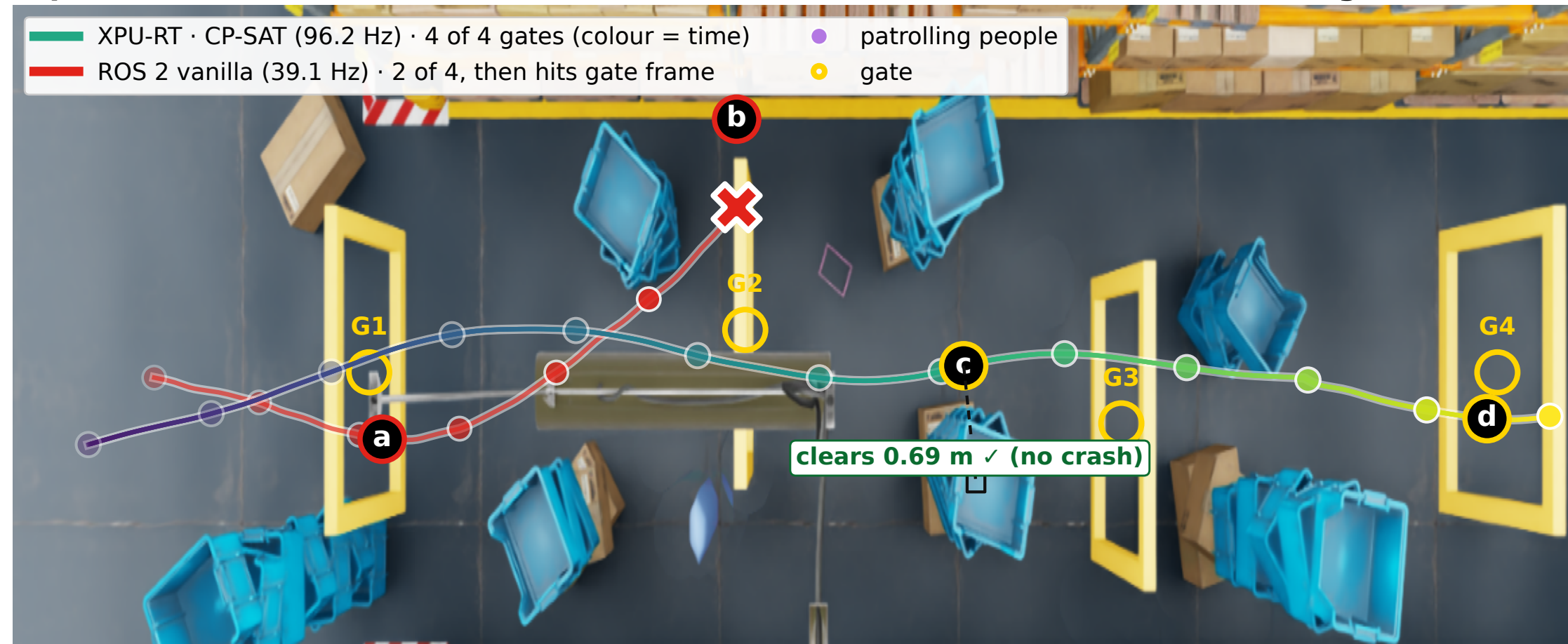
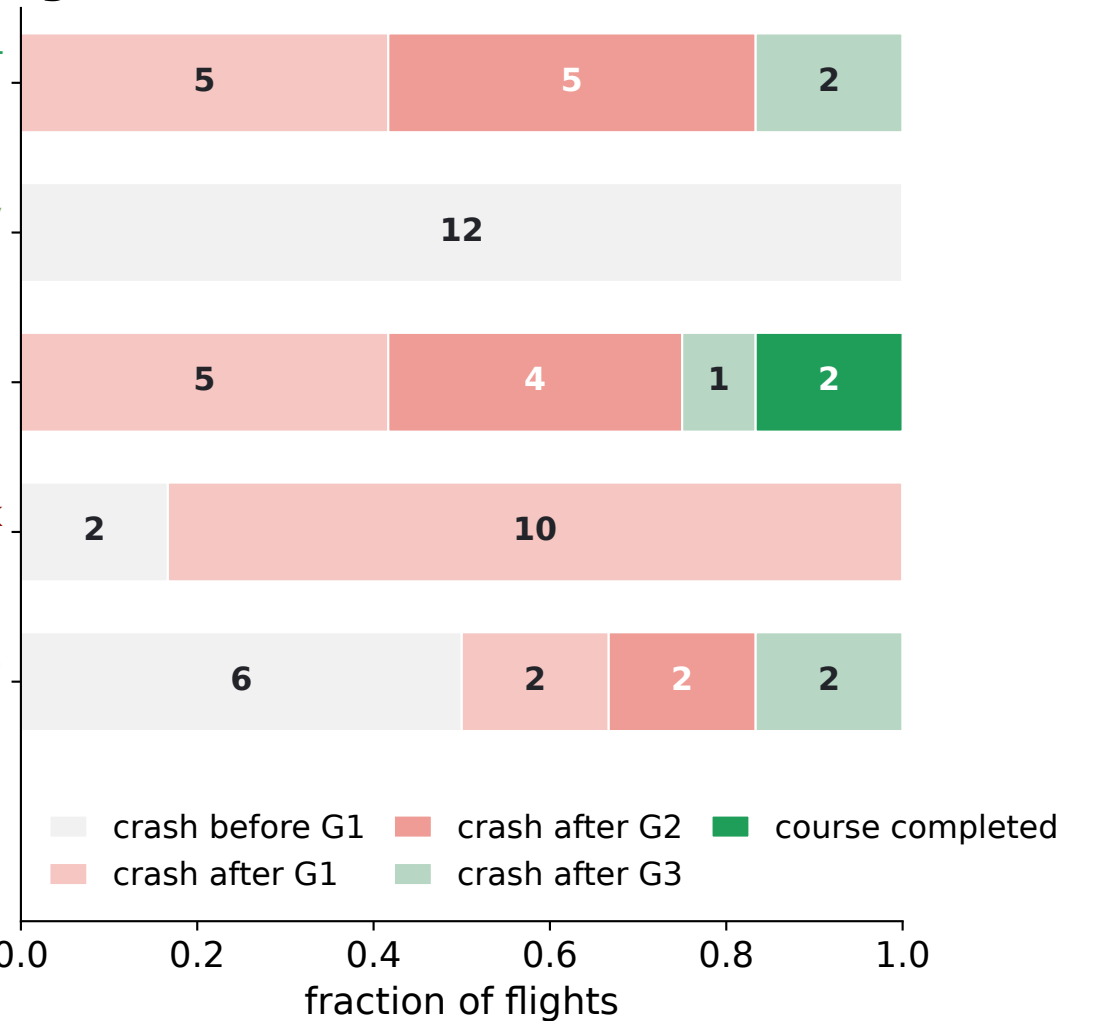


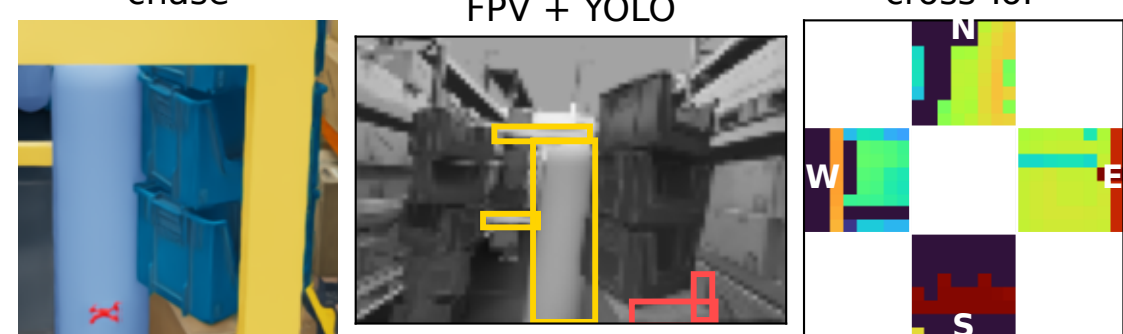
Warehouse gate course on the K1 — same scene and controller for both runtimes at 1.4 m/s; only what the board delivers differs — control every 10.4 vs 25.5 ms, camera→control latency and goal cadence, replayed



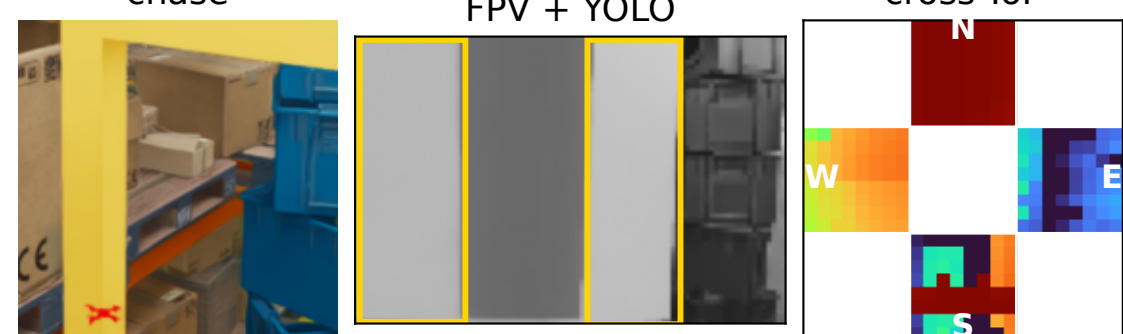
How far each arm gets at 1.4 m/s (gates cleared before the collision)



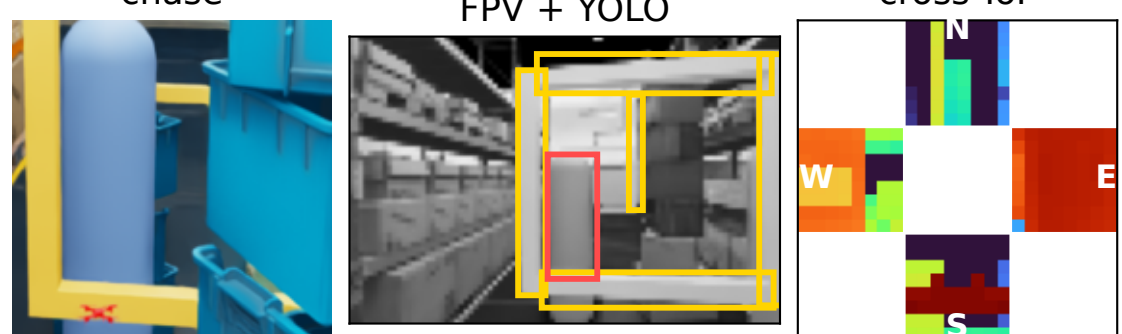
a ROS 2 39 Hz · clears G1 · t=2.6s



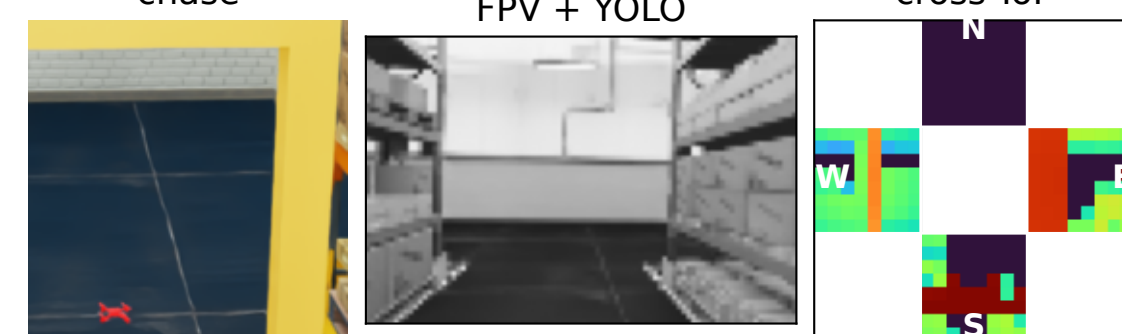
b ROS 2 39 Hz · hits gate frame after G2 · t=5.5s



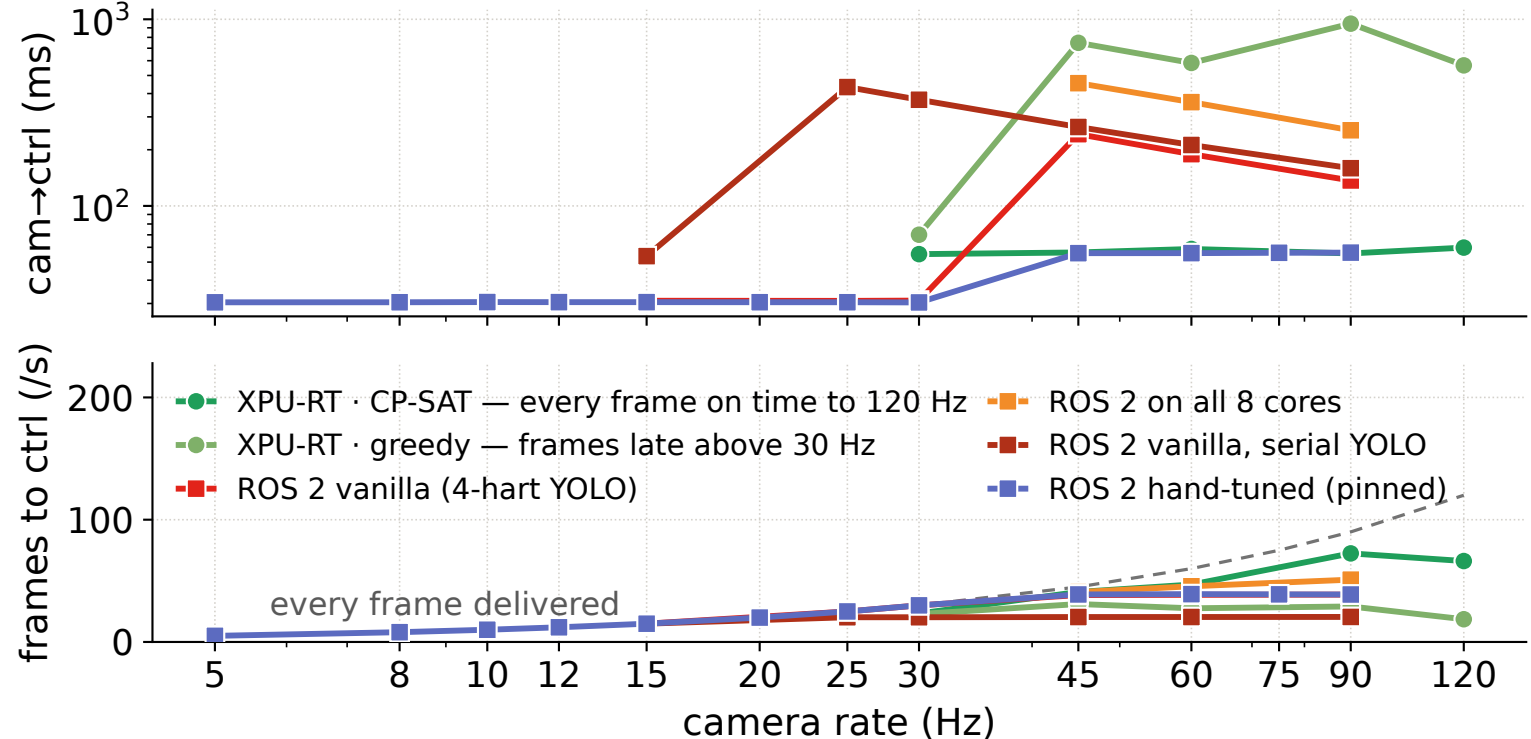
c XPU-RT 96 Hz · clears crate 0.69 m · t=7.4s



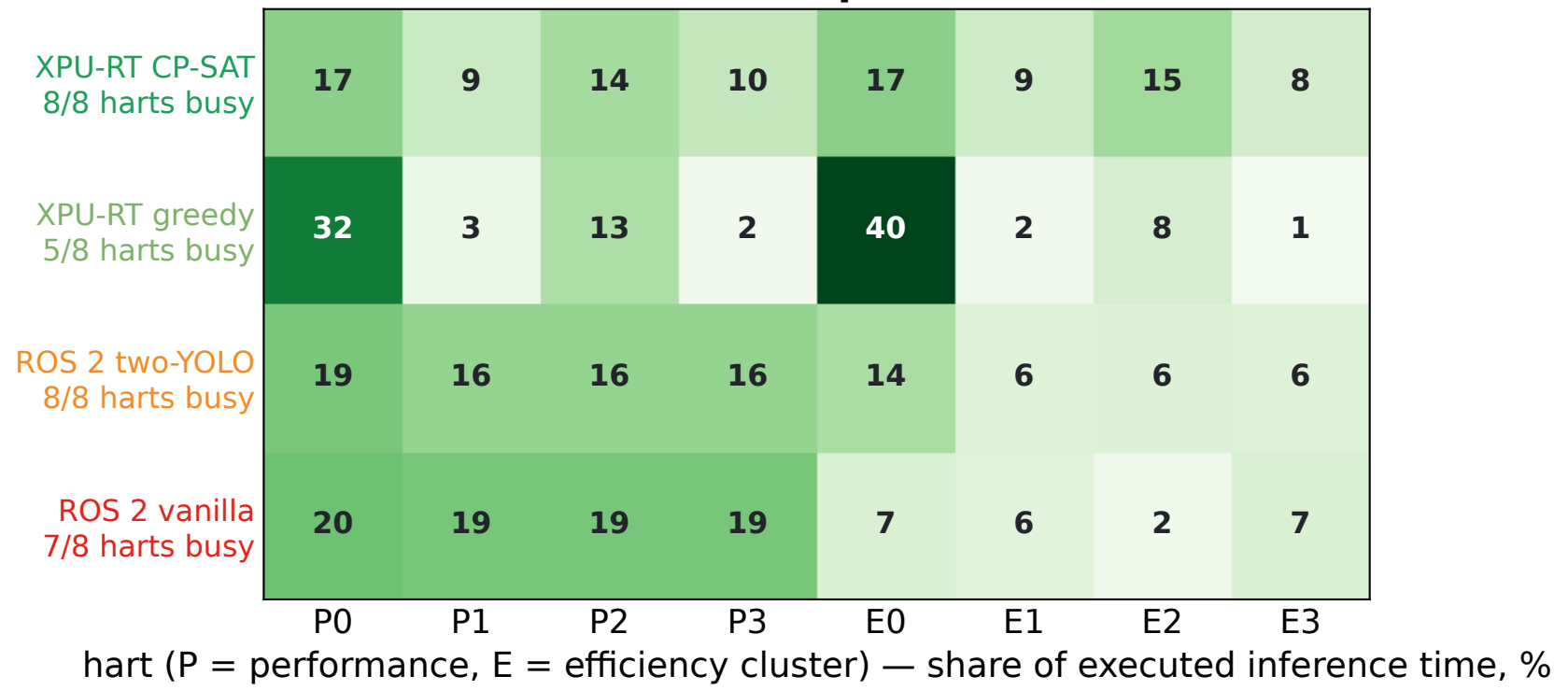
d XPU-RT 96 Hz · reaches gate · t=11.3s



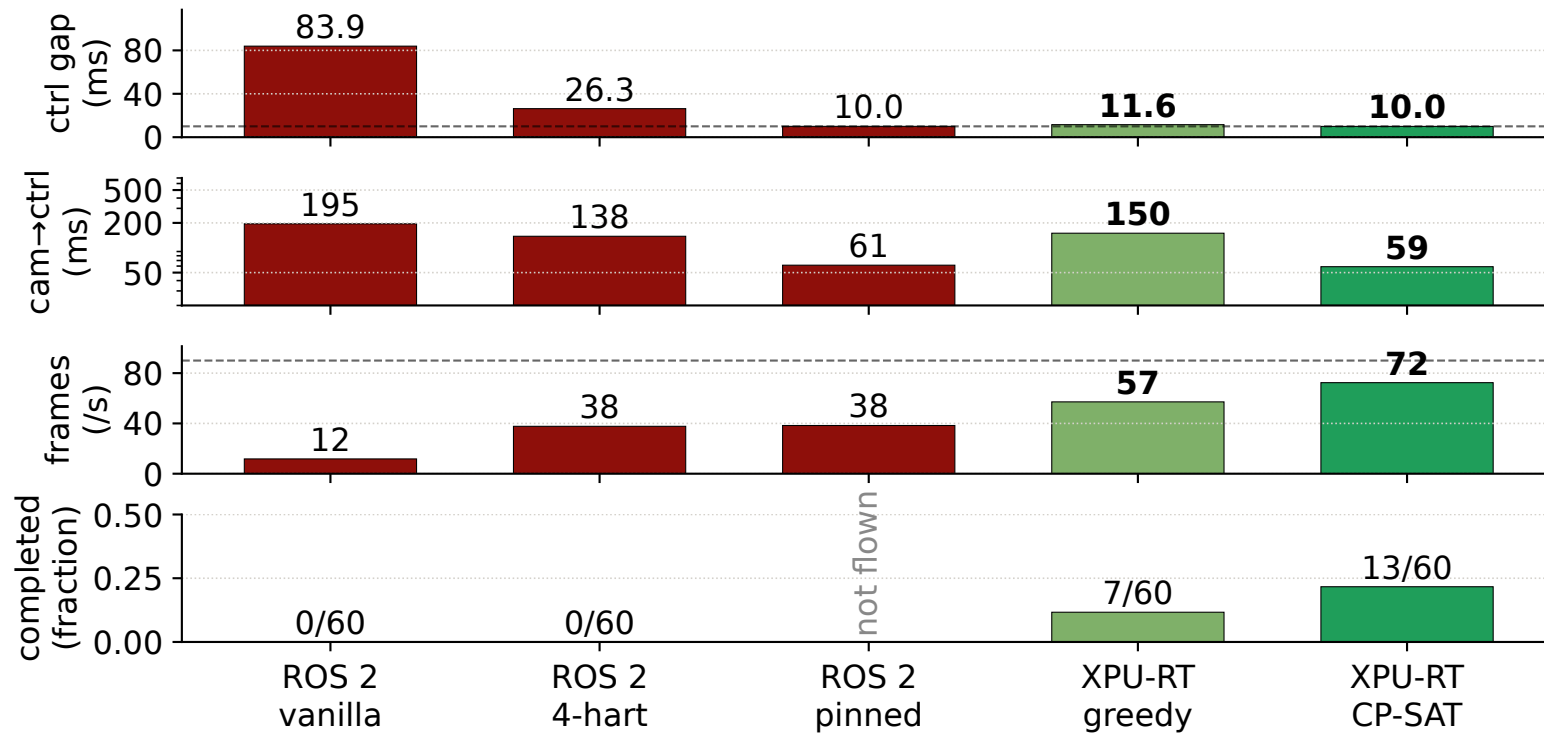
D Camera rate on the K1: camera→control latency and frames delivered, same kernels



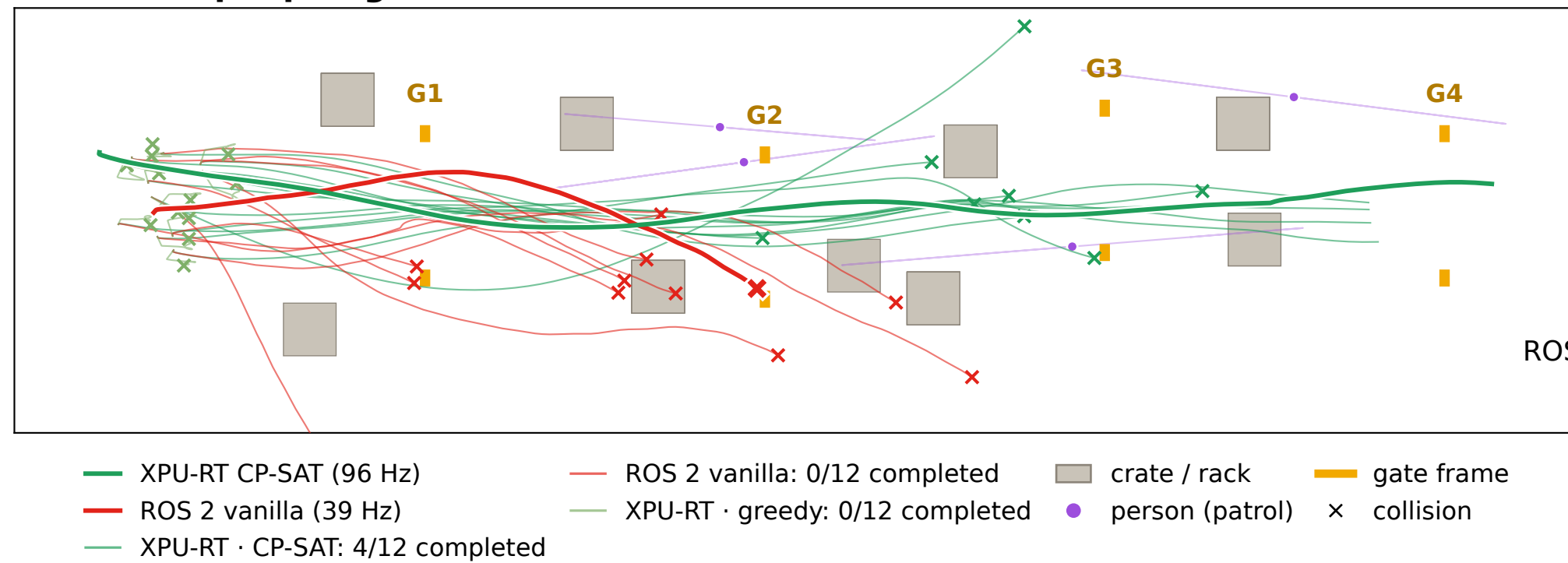
E Where the work lands, 45 Hz camera (board traces, three runs pooled)



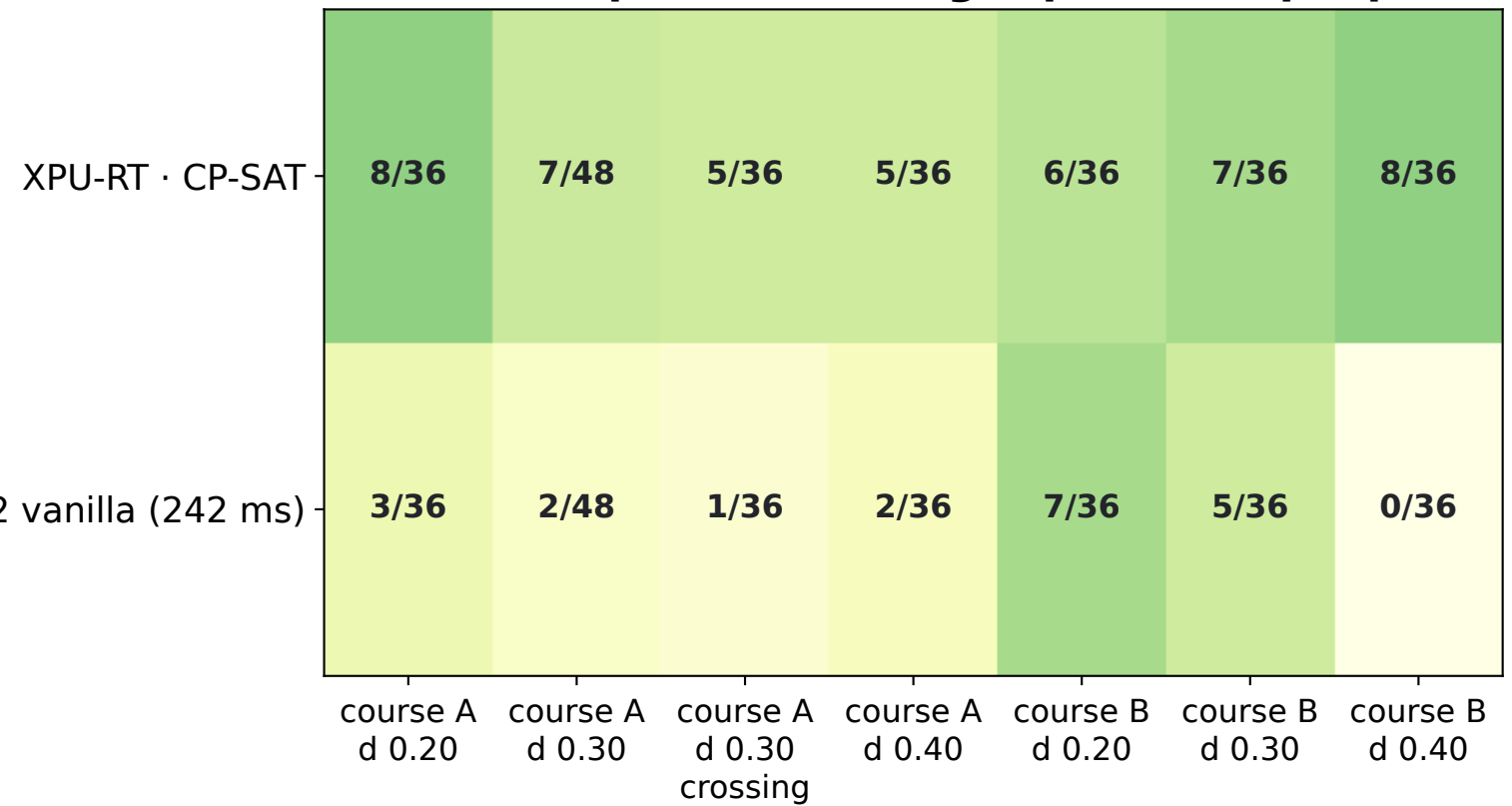
F Added load: 90 Hz camera + ffn_block + drone on the same 8 harts



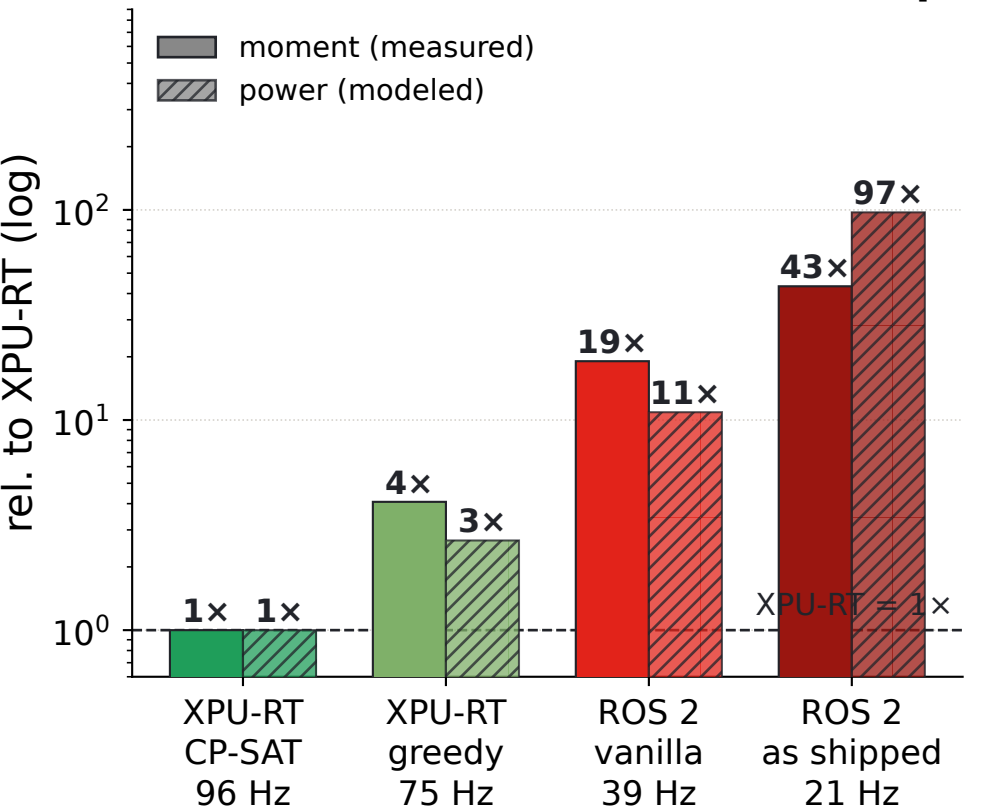
G The display scene, every recorded run (36 flights): obstacles, people, gate frames, collisions



H Environments both arms flew: course completed, 1.0–1.4 m/s pooled, one flight per seed, people 2.4 m



I Mechanism: the baseline thrashes (commanded moment, modelled power)



J Onboard K1, measured, 45 Hz camera — CP-SAT: 57 ms, every frame on time, control 10.0 ms · greedy: 748 ms, 44/44 late, control 13.2 ms

